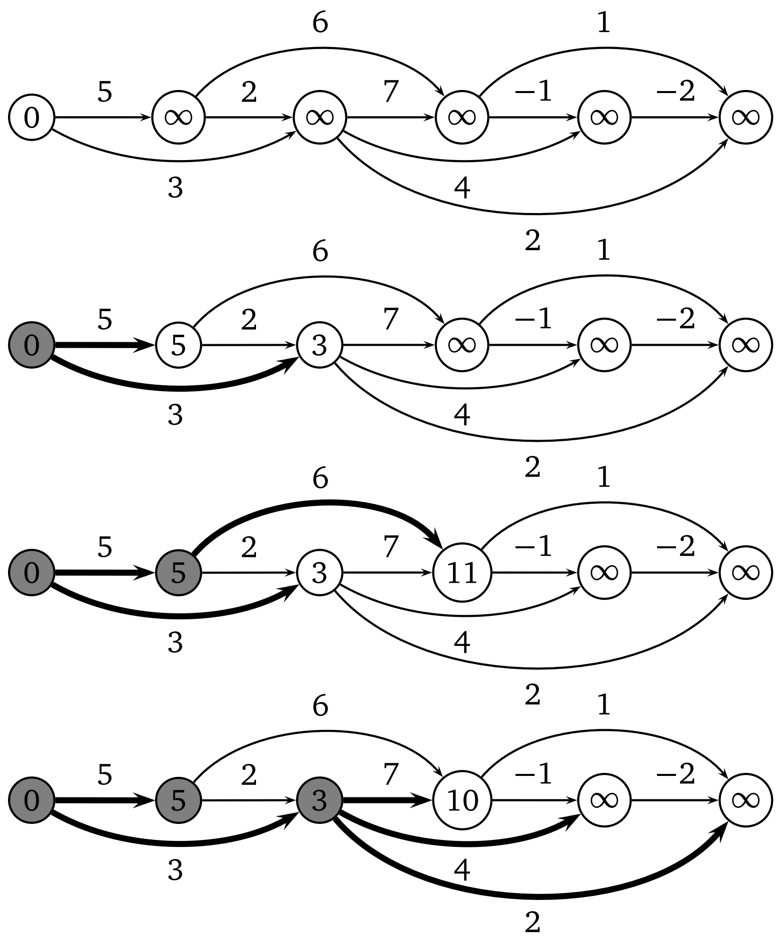
Introduction to Algorithms, Spring 2010
Homework 6 solutions

1 24.2-1



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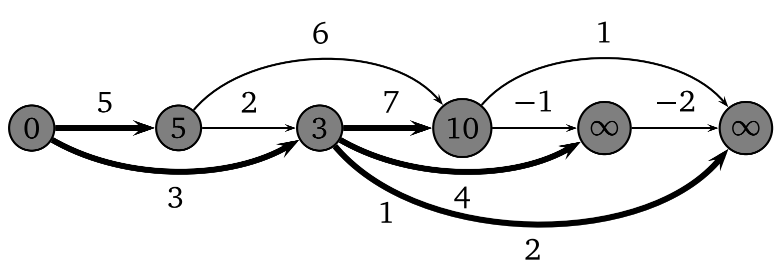


2 24.2-3

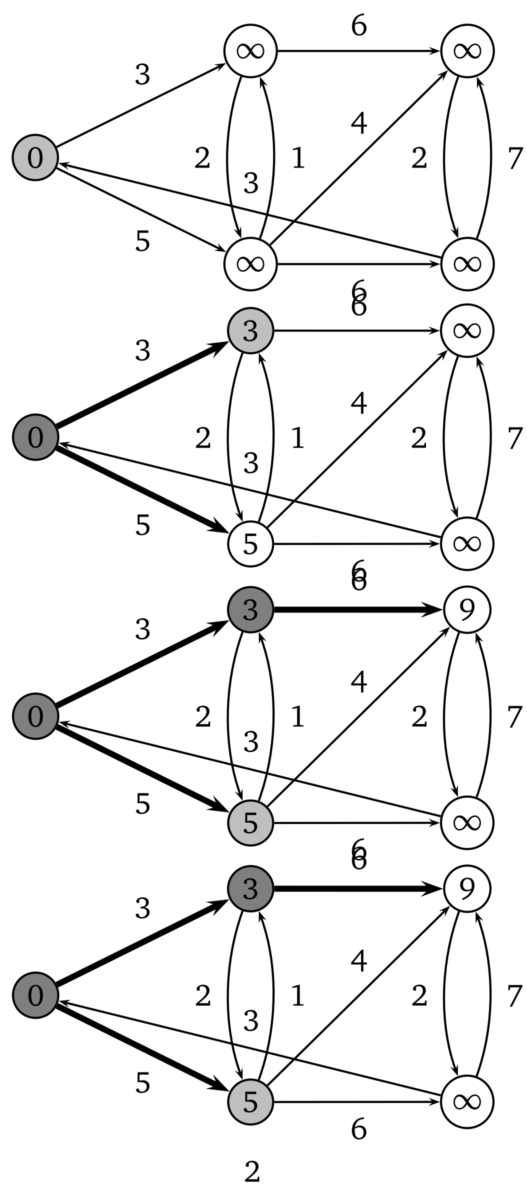
We modify the original graph $G = (V, E)$ as follows:

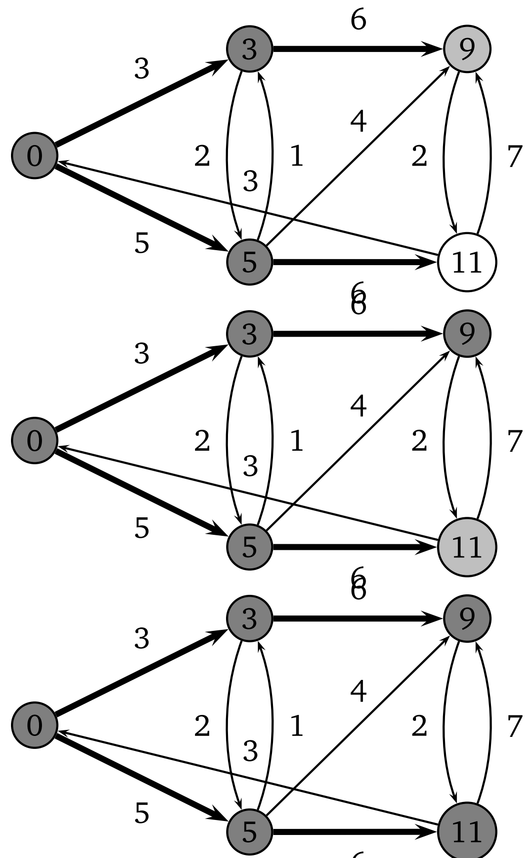
1. Add a source vertex s to V. For each u that the in-degree of u is zero, add an edge (s, u) to E.
2. For each edge $(u, v) \in E$, set the weight of (u, v) be the negation of the weight of v.

The above modification can be done in linear time. Then we apply DAG-SHORTEST-PATHS to the new graph.

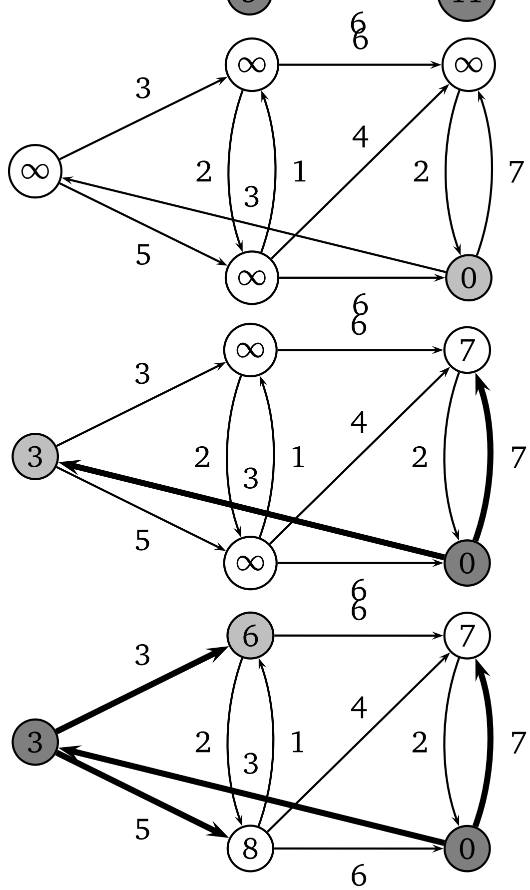
3 24.3-1

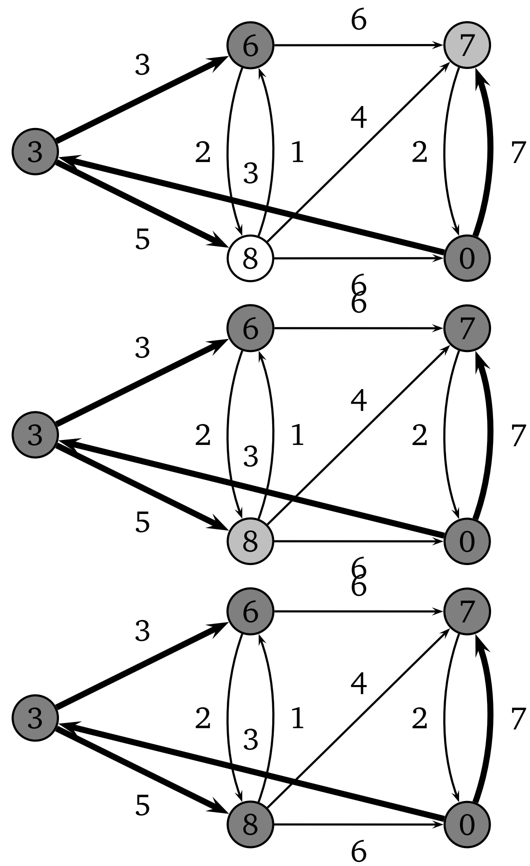
Source: s





Source: z





4 24.3-6

Here is the idea of this exercise. Because a real number $0 \leq r(u,v) \leq 1$ represents $0 \leq prob(u \rightsquigarrow v) \leq 1$, we can define $w(u,v) = -log_{10} r(u,v)$. Note that for $r(u,v) = 0$, we define $w(u,v) = \infty$, and for $0 < r(u,v) \leq 1$ we have $0 \leq w(u,v)$ (why?). Furthermore, since these probability are independent, the weight of path $p =< v_0, ..., v_k >$ is $w(p) = \sum_{i=1}^{k} w(v_{i-1}, v_i) = -log_{10}\{r(v_0,v_1)\cdots r(v_{k-1},v_k)\}$. This maps a communication channel problem to a directed graph with nonnegative weights, so we can apply Dijkstra's algorithm.

5 25.2-4

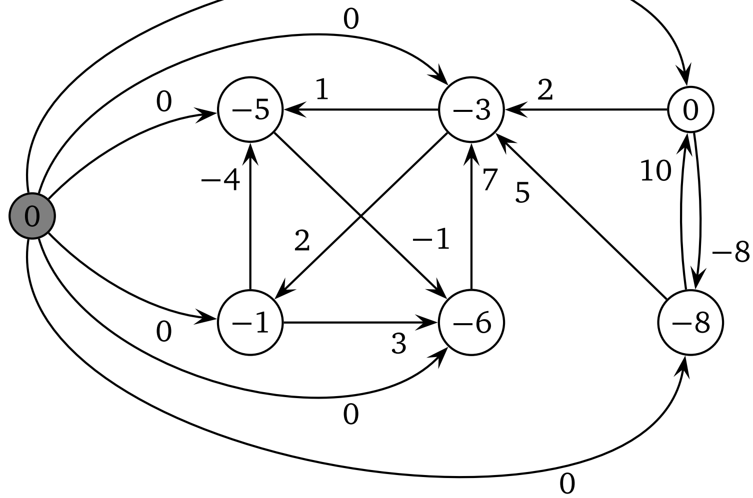
By definition, we have

$$d_{ik}^{(k)} = \min(d_{ik}^{(k-1)}, d_{ik}^{(k-1)} + d_{kk}^{k-1}) = \min(d_{ik}^{(k-1)}, d_{ik}^{(k-1)} + 0) = d_{ik}^{(k-1)}.$$

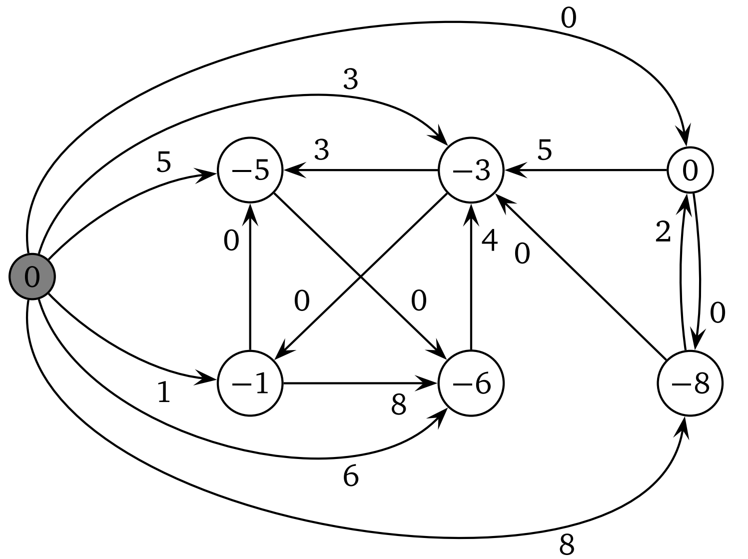
Similarly, we have $d_{kj}^{(k)} = d_{kj}^{(k-1)}$. Hence when computing d_{ij} at the kth iteration, the value d_{ij} will be the same as in the original algorithm even if d_{ik} or d_{kj} has been updated.

6 25.3-1

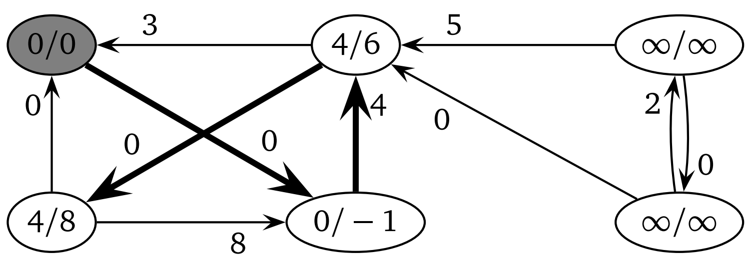
h values are in the vertices.



Weight $\hat{w}$



Shortest path started from 1



Shortest paths started from other vertices are similar.

7 25.3-5

Assume $c = \{v_0, v_1, \ldots, v_k, v_0\}$ and $p = \{v_0, v_1, \ldots, v_k\}$. Since c has 0-weight, $w(p) + w(v_k, v_0) = 0$. By the definition of $\hat{w}$, we have

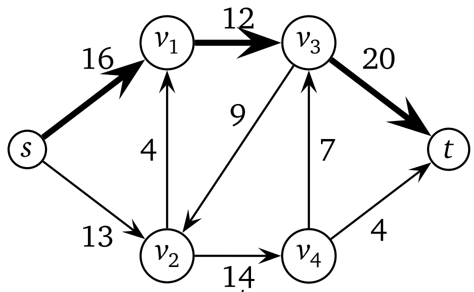
$$\hat{w}(p) = w(p) + h(v_0) - h(v_k) \tag{1}$$

$$\hat{w}(v_k, v_0) = w(v_k, v_0) + h(v_k) - h(v_0) \tag{2}$$

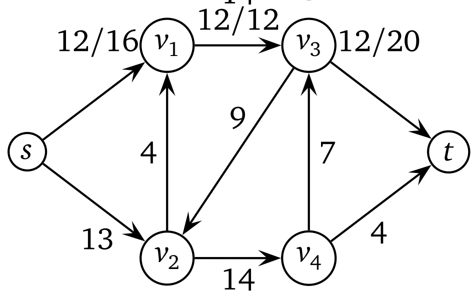
By combining the two equations above, we have $\hat{w}(p) + \hat{w}(v_k, v_0) = 0$. Since $\hat{w}(u,v)$ is nonnegative for all edges (u,v), $\hat{w}(u,v) = 0$.

8 26.2-3

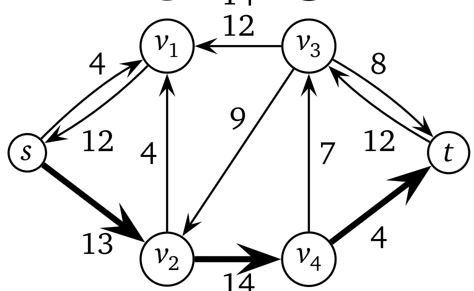
residual



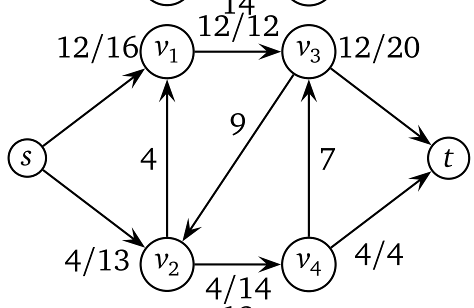
flow



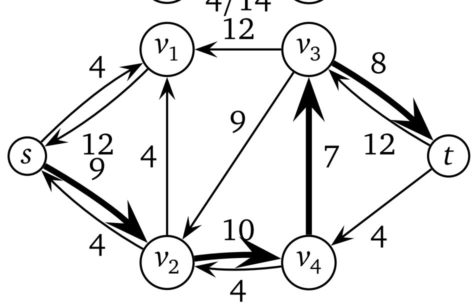
$\Longrightarrow$ residual



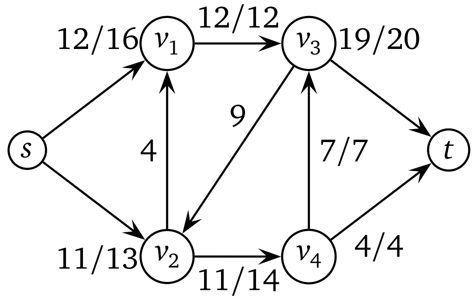
flow



$\Longrightarrow$ residual

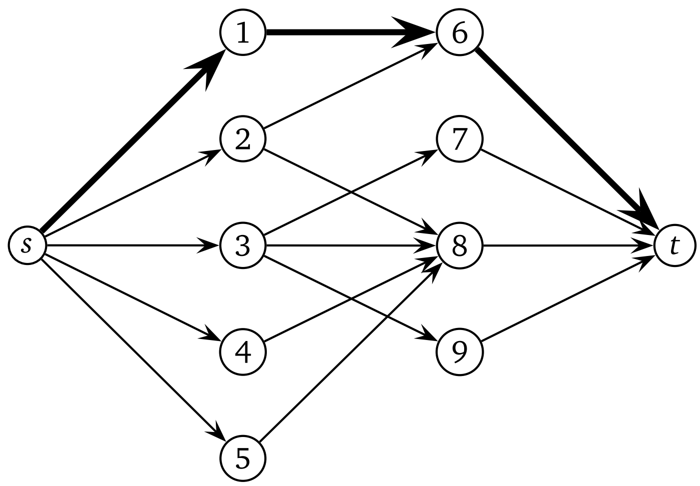


flow

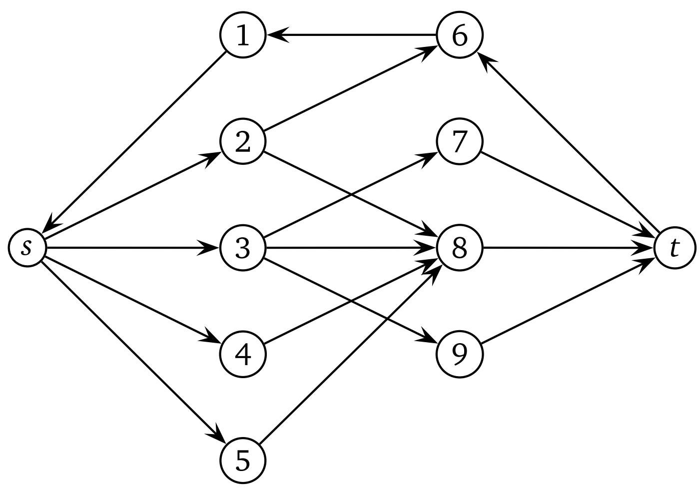


9 26.3-1

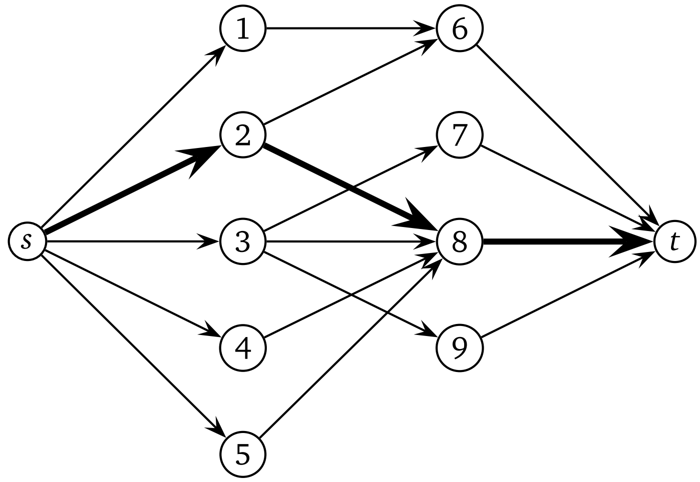
Step 1. $s \to 1 \to 6 \to t$



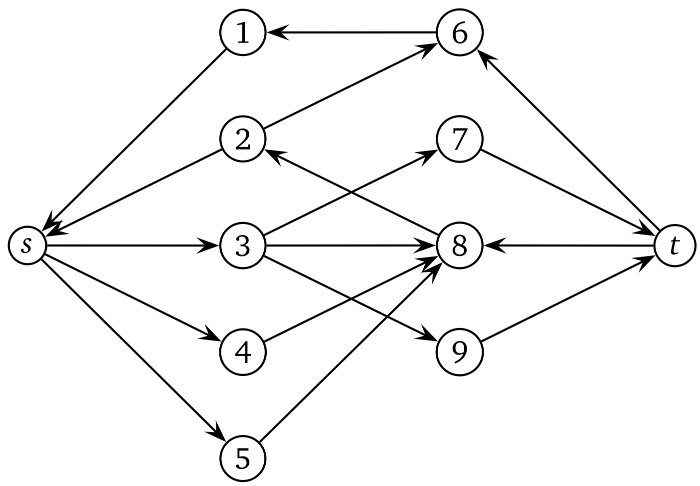
Residual



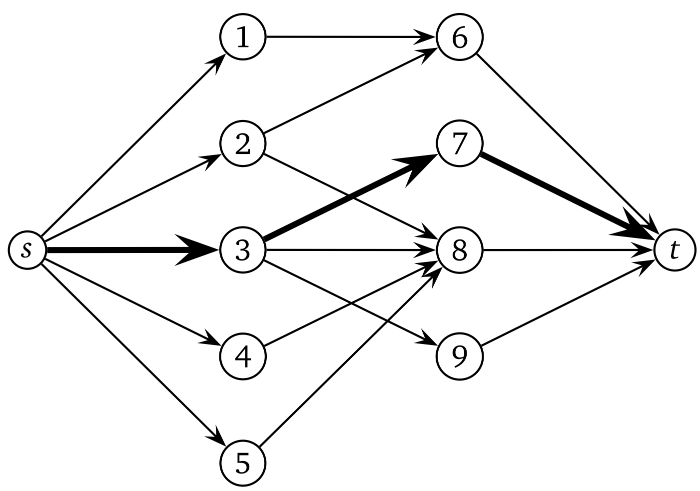
Step 2. $s \to 2 \to 8 \to t$



Residual



Step 3. $s \to 3 \to 7 \to t$



Residual